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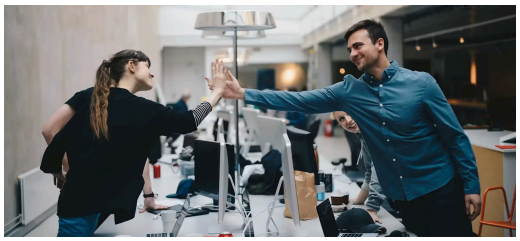
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Quality Assurance Engineer: Duties, Salary, and Top Skills

Written by Coursera Staff • Updated on Jul 30, 2026
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Quality assurance (QA) engineers work with multiple teams to ensure that a product meets quality requirements before its release. Discover seven top skills you need and learn how you can develop them to succeed in this role.



Key takeaways

- A quality assurance (QA) engineer monitors all phases of a project, identifying and resolving any issues before product launch.
- Quality assurance engineer specialist jobs have varying responsibilities, including creating test plans and test cases, reviewing technical document designs, setting and tracking QA metrics, and more.

- According to the US Bureau of Labor Statistics (BLS), the median quality assurance engineer salary is $102,610 [1].

To become a quality assurance engineer, you'll want to develop key skills, such as communication, critical thinking, and data analysis. Learn more about quality assurance engineers, along with seven essential skills you'll need to develop to prepare for a career in the field. Afterward, if you would like to build job-relevant skills, consider enrolling in Rice University's Engineering Project Management Specialization. In as little as four weeks, you'll have the opportunity to learn about engineering and stakeholder management, quality assurance, cost estimation, and more.



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What is a quality assurance engineer?

A quality assurance (QA) engineer monitors all phases of a new product or software to ensure it meets all requirements and standards. In this role, you will check for any bugs in the product, make sure any fixes are rectified before launch, and create test plans and test cases to ensure the accuracy of the product.

QA engineers usually have at least a bachelor's degree, along with experience in software development and knowledge of QA testing. In this role, you should also possess management skills to create positive relationships with other team members to help them effectively do their jobs.

What does a quality assurance engineer do?

Quality assurance engineers ensure that products meet quality standards throughout their development lifecycles. While their exact duties vary from one organization to another, common responsibilities include:

- Create test plans and test cases.
- Review technical design documents and provide feedback.
- Identify, document, and resolve bugs.
- Develop testing processes to ensure quality.
- Investigate the cause of bugs and other possible problems.
- Set and track quality assurance metrics.
- Continue to learn about new QA testing tools, strategies, and protocols.

Quality assurance engineer salary and job outlook

Quality assurance engineers earn a higher-than-average salary and are in demand.

According to the US BLS, software quality assurance analysts and testers earned a median salary of $102,610 as of May 2024, which is considerably higher than the median salary of $49,500 for all occupations in the country during the same period [1]. Furthermore, between 2024 and 2034, the US BLS projects that the number of job openings for QA professionals will grow by 10 percent, more than three times the job growth of just 3 percent for all occupations in the country [2].



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Quality assurance engineer skills

Quality assurance engineers need to be well-versed in their field and possess expertise in a variety of skills. Top skills QA engineers need include the following:

1. Communication

Being able to communicate can impact the project's success and your team's productivity. Working on a QA team requires continuous communication with everyone involved in the project, each step of the way. It's important to express your expectations clearly in simple terms and to match your listener's knowledge so that it's comprehensible to them.

Part of communicating is listening, which is an essential QA engineer skill. When you're working in a team, listening to them is imperative. After all, collaboration with others can reveal suggestions or challenges, ultimately contributing to the project's success. Listening to the thoughts and feedback from your development team, managers, and stakeholders can prove beneficial throughout the QA process.

Read more: What Is Effective Communication? Skills for Work, School, and Life

2. Critical thinking

Critical thinking allows you to assess and analyze the product during all its phases. This skill helps you recognize or predict any potential issues and come up with viable solutions. You can use your past experiences on previous projects to anticipate any problems with your current project. Instead of repeating mistakes, you can apply your knowledge and know-how from other projects to avoid pitfalls.

Critical thinking involves keeping an open mind, so discussing all possible issues and listening to solutions is essential. Then narrow down your choice of solutions to develop a systematic way to draw conclusions and foresee the value of the overall plan.

3. Data analysis

Data analysis is the practice of working with data to draw conclusions from its information. A QA engineer may use techniques like audits and documentation to gain important insights regarding product improvements.

To analyze data, you'll often create documentation listing all the testing processes included in the quality assurance objectives. Oftentimes, you'll work with a team, including stakeholders and developers, to create the testing strategy. It is crucial to have some familiarity with automated testing tools and knowledge of how to test products to find any potential flaws.



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4. Industry-specific knowledge

Industry-specific knowledge is imperative for those interested in QA engineer jobs, which many people choose to gain in bachelor's degree programs, boot camps, and other educational programs. Some companies may require a candidate with several years of experience in manual and automated software testing, testing analysis tools, product development tools, and QA methodologies. Having a strong understanding of technical skills, testing tools, and the company's QA process allows you to start working with less training.

Do I need to know programming languages?

Knowledge of specific QA programming skills is also essential. SQL is a language that QA engineers may need to know because it verifies all inserted, updated, or deleted values in the database and alerts you to any errors after processing. This tool works well for back-end testing. Other programming languages, including Java, JavaScript, C#, and Python, can also be helpful to know.



5. Leadership

As a QA engineer, leadership ensures that all team members understand and can complete their tasks. You should provide each team with the skills they need to complete their duties, including understanding the project's scope, meeting specifications, and identifying and rectifying any problems.

As a QA engineer, you may be in charge, but you must also be a team player. Maintaining a good relationship with other team members allows them to feel comfortable receiving any feedback you have to help them learn and improve. Creating a supportive and open environment where your team members can share ideas and feedback is vital.

6. Problem-solving

Because you'll be tracking bugs and other issues, you need to use your problem-solving skills to anticipate how to resolve these in advance. First, you should analyze the problem so you can devise a viable solution. It's

important to use problem-solving skills to figure out what leads to the problem and how you can come up with a permanent solution to ensure the same problem doesn't occur in the future.

7. Time management

As a QA engineer, you'll often work on tight deadlines, making time-management skills crucial. Planning ahead and prioritizing your day may help you be more productive and reduce unnecessary stress. In addition to your workload, you are also responsible for ensuring that teams meet their deadlines. If one area of the process falls behind or makes a mistake, the timeline affects other QA teams, so staying on top of projects is essential for keeping the team moving along.

Some effective time-management methods include:

- Learning how to delegate tasks
- Getting organized and tracking your time using software and other time-management tools
- Conducting efficient meetings that end with action items and deadlines.

Discover free career-planning resources

Explore our Career Resource Hub to learn more about career paths, assess your skills, and take a quiz. You can also access the following free resources to learn more key topics in QA engineering:

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